Active Maintenance Report: coppock_mcclellan_2019

Coppock and McClellan (2019, Research & Politics): Validating the Demographic, Political, Psychological, and Experimental Results Obtained from a New Source of Online Survey Respondents

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Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Coppock and McClellan (2019), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1177/2053168018822174
Replication archive	10.7910/DVN/DDWWJW
Pre-analysis plan	None. The article states that the analyses were not pre-registered.

The data are not redistributed here. The deposit is 29 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it and holds the extraction of every claim the article makes. `errata.qmd` builds `coppock_mcclellan_2019_errata.pdf`, which lists the sentences in the article that the deposited data do not support. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_mcclellan_2019.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 29 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, broom, scales, matrixStats, rsample, sandwich, modelsummary, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory and the scripts run equally well under `Rscript` outside RStudio. Individual scripts can be run on their own in any order, with one exception: `text_in_text_claims.R` reads the table outputs, so it comes after them.

The full run takes about nine minutes, eight of them in the two scripts that repeat the appendix's bootstrap two hundred times each to measure how much of it is noise. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV, TeX and PNG output should come back byte-identical. The five PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

On this machine, almost. Eleven of the twelve numbered scripts run to completion on a current R installation, and every package they name still installs from CRAN, `coefplot` and `matrixStats` included. The one script that fails is `06_KamSimasReplicationAnalysis.R`, and it fails inside `stargazer`, on a bug that turns on how long the names of its model objects are.

On a case-sensitive filesystem, no. Eleven of the archive's twelve `load()` calls name a file the deposit does not contain: the code asks for `WelfareReplicationStacked.rdata` and Dataverse holds `WelfareReplicationStacked.RData`. macOS and Windows resolve the mismatch silently and Linux does not, so ten of the thirteen scripts halt on their first data line for anyone who downloads the deposit onto a case-sensitive disk. Details on both below.

The failure lands badly. The published article prints one table, a prose summary of the five experiments, and its online appendix prints two, the distance tests and the Kam and Simas probit specifications. Script 06 is the one that makes the probit table, so exactly one of the paper's two numeric tables sits behind the only script that does not run.

And whether it runs is a smaller question than it sounds, because the deposit writes nothing. Across all thirteen scripts the only uncommented call that touches a file is one `source()`. Every `ggsave`, `sink`, `write.csv` and `print.xtable` is commented out, several under the heading `# Uncomment to save`. Running all twelve numbered scripts in a scratch copy of the deposit leaves every file in that copy byte-identical to the deposited version: nothing is created, nothing is overwritten.

Most of the scripts do not even print. Scripts 01, 02, 03 and 09 assemble their table into an object on the last line and stop, so the four descriptive tables the archive's own `README.txt` advertises never reach the screen either. Scripts 10 and 11 do the same with their plot objects. What a full run actually emits is four LaTeX tables on standard output, from scripts 04, 05, 07 and 08, and one line of counts from script 12. So there is no artifact a reader can compare against the published paper, and the eleven scripts that run clean cannot be said to reproduce anything. Reading the numbers back out means capturing them from the session rather than from files.

32 of the 199 published claims the deposited scripts can be checked against fail to reproduce. 25 of them are standard errors in appendix Table 2, and none is a coefficient. 5 of the remaining 7 are places where the paper disagrees with itself, and the last is a bootstrap standard error the deposit's unseeded procedure cannot pin down. See the errata below.

1.2 Does the maintained rewrite reproduce the paper?

Yes. 192 of the 199 verifiable ground truth claims match the published values to reported precision. 5 of the 7 that do not are places where the article disagrees with its own appendix rather than with the code: a sample size the appendix appears to have copied from a different experiment, a sample size the appendix's prose and its own table state differently, and a count of significant differences that the article gives as five where the appendix table shows six. The last is a bootstrap standard error whose published value falls just outside the range the deposit's own unseeded procedure produces, discussed under bootstrap stability below. The remaining 60 recorded quantities are ones the rewrite computes but the paper never states.

Two published quantities turn out not to mean what the paper says they mean. The p-values in appendix Table 1 are described as "two-tailed p-values under a normal approximation" and are not: the archive computes `pnorm(2 * (1 - |z|))`, which calls a difference significant at the 0.05 level once `|z|` passes 1.82 rather than 1.96. Separately, the deposited probit script omits the `se = starprep(...)` argument that the OLS call in the same file carries, so as deposited it would print model-based standard errors where the published table carries the HC2 robust ones its note describes. The rewrite reports both quantities in both cases and changes nothing silently.

A third finding is about precision rather than correctness. The appendix's headline count, that Lucid is significantly closer to the 2012 ANES than MTurk on 14 of 21 characteristics, rests on standard errors from 100 bootstrap resamples drawn without a seed. Running the deposited procedure 200 times returns 14 on 40% of them and ranges from 12 to 15. The number is not wrong; it is simply not pinned down by the deposit.

2 Paper overview

Citation: Coppock, A. and McClellan, O. A. (2019). "Validating the demographic, political, psychological, and experimental results obtained from a new source of online survey respondents." *Research & Politics*, 6(1). DOI: 10.1177/2053168018822174

Summary: The paper asks whether the Lucid Fulcrum Exchange is a usable source of subjects for survey experiments, following the template Berinsky, Huber and Lenz (2012) used for Mechanical Turk. A single quota-sampled Lucid sample of 3,504 respondents collected in March 2016 is compared with MTurk, the 2012 ANES, the CPS, the ANES panel, the CCES and the CCAP on demographics, political attitudes and Big Five traits, and five experiments are replicated on it: the GSS welfare question-wording experiment, Tversky and Kahneman's Asian Disease problem, Kam and Simas (2010) on risk acceptance and framing, Hiscox (2006) on free-trade frames, and Berinsky (2017) on death-panel rumor corrections. Lucid is closer to the 2012 ANES than MTurk on 18 of 21 standardized characteristics, and four of the five experiments recover treatment effects close to the originals. The exception is the rumor-correction experiment, where none of the corrections works as well on Lucid in 2016 as it did on the original sample in 2010.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0. Scripts were run from a scratch copy of the deposit stripped to code and data, and the deposit's checksums were re-verified afterwards.

Script	Status on current R	Resolution
LucidValidationHelperFunctions.R	Clean (sourced by nine scripts)	No changes required
01_DemographicsTableAnalysis.R	Runs; funs() and melt/dcast deprecation warnings	No changes required
02_PoliticalTableAnalysis.R	Runs; same warnings	No changes required
03_PsychologicalTableAnalysis.R	Runs; same warnings	No changes required
04_WelfareReplicationAnalysis.R	Runs	No changes required
05_AsianDiseaseReplicationAnalysis.R	Runs	No changes required
06_KamSimasReplicationAnalysis.R	FAILS at the first stargazer call	Assign the filtered subsets before fitting, or drop stargazer
07_HiscoxReplicationAnalysis.R	Runs	No changes required
08_HealthRumorsReplicationAnalysis.R	Runs	No changes required
09_SurveyBehaviorTableAnalysis.R	Runs; same warnings	No changes required
10_StandardizedPlots.R	Runs; geom_errorbarh() deprecation warning	geom_linerange(); position_dodge() on the discrete axis
11_StandardizedPlotsHeterogeneousEffects.R	Runs; same warning, plus duplicated scale warnings	Same, and keep one scale per aesthetic
12_DistanceTests.R	Runs; funs() warning and unnest() without cols=	unnest(cols = c(model))

Nothing here is a missing package. `matrixStats`, `coefplot`, `stargazer`, `xtable`, `reshape2` and `estimatr` all install from CRAN today. `geom_errorbarh()` is deprecated in ggplot2 4.0 but still draws, so scripts 10 and 11 produce their figures with a warning rather than an error, and `coefplot::position_dodgev()` is loaded by both of them and still works.

The "Runs" in that table means the script reaches its last line without an error. It does not mean the script produced anything, because none of them does: the deposit's every write is commented out, so a complete run of all twelve leaves the directory exactly as it found it. That is worth separating from the two defects above, since a script can pass a "does it run" check while being unable, in principle, to produce a file anybody could check.

3.1 The stargazer failure

`06_KamSimasReplicationAnalysis.R` fits nine models and passes all nine to `stargazer`. To label the columns, `stargazer` recovers the names of the objects it was given:

```
object.names.string <- deparse(substitute(list(...)))
.global.object.names.all <- .get.object.names(object.names.string)
```

`deparse()` breaks its output into several strings once the deparsed expression passes 60 characters, so `object.names.string` is a character vector rather than a scalar whenever the argument names are long enough. `.get.object.names()` hands it straight to `.inside.bracket()`, which opens with

```
if (!is.character(s)) return("")
if (is.null(s)) return("")
if (is.na(s)) return("")
if (s == "") return("")
if (length(s) > 1) return("")
```

The guard for a vector argument is the last of the five, three lines after the `is.na()` that a vector argument breaks. Until R 4.2 a length-greater-than-one condition in `if` was a warning and the function limped through; R 4.2 made it an error, and the script now halts with `the condition has length > 1`.

Nothing about the failure involves `filter()`, the `data =` argument, or the `==` inside it. `list(fit_kam_lucid, fit_kam_lucid_ctrl, ... fit_kam_kam_X)` deparses to three lines; the argument lists of scripts 04, 05, 07 and 08 have between two and four models with shorter names and deparse to one line each, which is the whole of the difference. Passing the same nine fitted models under one-character names produces the table without complaint. Pre-assigning the filtered subsets before fitting does not help, because it leaves the nine long argument names exactly where they were.

The consequence is larger than one table. The probit models of appendix section 2.3.4, and the second `stargazer` call that formats them, sit below the failing line, so nothing after it runs.

3.2 The data file names

The deposit's data files are named `.RData` with one exception, `DemosTableStacked.rdata`. The scripts ask for `.rdata` without exception. The result is eleven `load()` calls across ten of the thirteen scripts that name a file the deposit does not contain, and one call, in `01_DemographicsTableAnalysis.R`, that happens to be right.

Table 2: The eleven load() calls whose file name does not match the deposit. Each runs on macOS and Windows and stops on Linux.

Script	Name requested	Name deposited
02_PoliticalTableAnalysis.R	PoliticsTableStacked.rdata	PoliticsTableStacked.RData
03_PsychologicalTableAnalysis.R	PsychTableStacked.rdata	PsychTableStacked.RData
04_WelfareReplicationAnalysis.R	WelfareReplicationStacked.rdata	WelfareReplicationStacked.RData
05_AsianDiseaseReplicationAnalysis.R	AsianDiseaseStacked.rdata	AsianDiseaseStacked.RData
06_KamSimasReplicationAnalysis.R	KamSimasReplicationStacked.rdata	KamSimasReplicationStacked.RData
07_HiscoxReplicationAnalysis.R	HiscoxReplicationStacked.rdata	HiscoxReplicationStacked.RData
08_HealthRumorsReplicationAnalysis.R	HealthRumorsReplicationStacked.rdata	HealthRumorsReplicationStacked.RData
09_SurveyBehaviorTableAnalysis.R	SurveyBehavior.rdata	SurveyBehavior.RData
10_StandardizedPlots.R	StandardizedDemos.rdata	StandardizedDemos.RData
10_StandardizedPlots.R	StandardizedExperiments.rdata	StandardizedExperiments.RData
11_StandardizedPlotsHeterogeneousEffects.R	standardizedExperimentsHeterogeneousEffects.rdata	standardizedExperimentsHeterogeneousEffects.RData

The deposit's own `README.txt` writes the same twelve names the same wrong way, ending each in `.rdata`, and gets only `standardized_stacked_st.rds` right. So the code and the document that describes it agree with each other and both disagree with the files. `README.txt` gets two other names wrong too. It lists `11_StandardizedExperimentsHeterogeneousEffects.R`, and the deposited file is `11_StandardizedPlotsHeterogeneousEffects.R`. It assigns "Appendix Table 1" to two different scripts, 06 and 12, and only one of them can be right: script 12 makes appendix Table 1 and the probit half of script 06 makes appendix Table 2.

This class of defect cannot be detected by running the archive on the machine it was written on, and it is invisible to a checksum, which verifies the bytes of a file and says nothing about the name a script uses to ask for it. The maintained rewrite spells every one of these names exactly as the deposit does, and none of the deposited files was renamed to accommodate it.

3.3 What the archive says it produces, and what the paper prints

The deposited `README.txt` maps its scripts onto nine numbered tables:

> `01_DemographicsTableAnalysis.R` (Produces Table 1) … `09_SurveyBehaviorTableAnalysis.R` (Produces Table 9)

No such tables appear in the published article or in the online appendix. The article's only table is a prose summary of the theoretical scope of the five experiments; the appendix has two, the distance tests of section 1 and the probit specifications of section 2.3.4. The nine numbered tables belong to an earlier draft, and the article's own text still points at one of them: "included in the online appendix, Table 2" is used twice, for political knowledge and for policy preferences, and the appendix's Table 2 is the probit table.

That leaves the demographic, political, psychological, welfare, Asian Disease, Kam and Simas OLS, Hiscox, Berinsky and survey-behavior results with no published form other than the standardized versions plotted in Figures 1 and 2 and a handful of numbers quoted in prose. All nine are computed here, under the archive's own numbering, and each script says in its header that the number is the archive's rather than the paper's. The prose numbers are checked one by one in `maintained/text_in_text_claims.R`.

Every `value_paper` in this repository's ground truth is transcribed from the article or the online appendix and from nowhere else. Quantities the paper does not state are left blank rather than filled from a script run, which is why the nine archive tables contribute rows to the ground truth without contributing comparisons.

4 Errata

Two published quantities do not come from the procedure the paper describes. Neither is a coding slip that the maintained rewrite silently corrects: both are reported in both forms, so the published number stays reproducible and the intended one is available beside it.

Seven sentences in the article and its appendix state something the deposited data do not support, and two further entries correct the published pages themselves, where the deposit has nothing to say. All nine are collected in `coppock_mcclellan_2019_errata.pdf` at the root of this repository, built by `errata.qmd`, which recomputes every number in every corrected sentence from the deposit each time it is rendered. None of the nine changes a conclusion. What follows here is the analysis behind them, numbered as the note numbers them, plus two further findings that could not go in an errata because the deposit does not settle what the corrected sentence should say.

4.1 The appendix Table 1 p-values

Section 1 of the online appendix says the standard errors come from a nonparametric bootstrap and that "we obtain two-tailed p-values under a normal approximation." The archive's `12_DistanceTests.R` computes

```
p = pnorm(q = 2 * (1 - abs(lucid_is_closer / se)), lower.tail = TRUE)
```

which is not a p-value for any hypothesis. A two-tailed normal-approximation p-value is `2 * pnorm(-abs(z))`. The two expressions agree in spirit and differ in threshold: the archive's calls a difference significant at 0.05 once `|z|` exceeds 1.82, where the two-tailed test needs 1.96. The published p-value column reproduces exactly under the archive's expression and not under the two-tailed one, so the published table is the archive's expression, not the appendix's description of it.

Table 3: Appendix Table 1 p-values under the archive's expression and under the two-tailed test the appendix describes. Bootstrap standard errors are stochastic, so the reproduced column will not agree digit for digit with the published one; see the stability section.

Variable	Published	Archive expression	Two-tailed
Female	0.000	0.000	0.004
Education	0.000	0.000	0.000
Age	0.000	0.000	0.000
Mean income	0.000	0.000	0.003
White	0.873	0.870	0.662
Black	0.000	0.000	0.000
Hispanic	0.357	0.383	0.251
Northeast	0.179	0.131	0.118
Midwest	0.409	0.398	0.259
South	0.010	0.003	0.018
West	0.948	0.948	0.850
Voter registration	0.157	0.177	0.143
Voter turnout	0.000	0.000	0.004
Party ID	0.010	0.011	0.032
Ideology	0.000	0.000	0.000
Political Interest	0.000	0.000	0.000
Extraverted	0.034	0.057	0.074
Agreeable	0.001	0.000	0.006
Conscientious	0.000	0.000	0.000
Stable	0.000	0.000	0.000
Open	0.005	0.002	0.016

What turns on it: the appendix reports that Lucid is significantly closer to the ANES than MTurk on 14 of the 21 characteristics. Read off the standard errors the published table prints, the two-tailed test gives 13: extraversion is the only one of the 21 whose verdict falls between the two thresholds, its published `z` of 1.92 sitting between 1.82 and 1.96. In the article, the sentence "Formal hypothesis tests demonstrate that Lucid is significantly closer to the ANES 2012 than MTurk on all five traits" becomes four of five. These are entries 1 and 2 of the note.

The published count is 14 and this repository's own bootstrap now returns 13 under the archive's expression as well as 13 under the two-tailed test, so the two corrections no longer depend on which expression is read. The reason the two numbers used to agree is that the maintained run then drew the deposit's 100 resamples: at that `B` the count is itself a coin

toss, reaching 14 on 39.5 per cent of 200 seeds. That is entry 7, and the next section is the measurement behind it.

4.2 The appendix Table 2 standard errors

The note under appendix Table 2 reads "Robust standard errors are in parentheses," and its numbers are HC2 sandwich standard errors, consistent with the article's statement that "In all cases, we estimate HC2 robust standard errors." The deposited probit `stargazer` call passes no `se =` argument, unlike the OLS call in the same script, so the code as deposited would print model-based probit standard errors instead. All 30 published coefficients, all nine sample sizes, all nine log-likelihoods and all nine AIC values reproduce exactly. 25 of the 30 published standard errors do not, and 30 of the 30 match once the robust variance the note describes is used.

The maintained rewrite reports both: `robust.std.error` reproduces the published table, `std.error` is what the deposited script would print.

4.3 Where the article disagrees with itself

None of these is an archive defect; all are visible only once the numbers are recomputed.

The article says that of the 11 demographic variables, "the Lucid mean is closer to the ANES mean in nine instances, five of which are statistically significant." Nine is right. The appendix table's own p-value column marks six of those nine significant, not five: female, education, age, mean income, black and south. Six is also the count under the two-tailed test, so this one does not turn on the p-value expression above.

"The regional balance on Lucid comes very close to the 2012 ANES, whereas the MTurk sample appears to overrepresent southerners" has the direction backwards. The southern share is 30.1 per cent on MTurk against an ANES 2012 benchmark of 37.2, and Figure 1, where the sentence sends the reader, puts MTurk 0.146 standard deviations below the benchmark. MTurk over-represents the Northeast and the Midwest.

Appendix section 2.1.3 gives the Lucid welfare replication a sample of 1,811, which is the Lucid sample size of the Hiscox experiment reported in section 2.4.2. The welfare data have 3,294 respondents. Appendix section 2.3.1 gives the original Kam and Simas study 761 respondents where appendix Table 2 reports 752 for the same models, and section 2.5.1 describes the original Berinsky sample as MTurk where the article describes it as Survey Sampling International.

4.4 Two findings the deposit does not settle

The article says the Lucid sample corresponds "more closely to the 2012 ANES baseline for voter registration and voter turnout," and says on the same page that "MTurk is significantly closer for voter turnout." Both statements are supported by a deposited file, and the two files disagree. The standardized file the distance tests read puts the ANES 2012 turnout benchmark at 0.702, which makes MTurk the closer sample and gives appendix Table 1 its −0.124; the political table's weighted mean puts the same benchmark at 0.756, which makes Lucid the closer sample. The registration comparison is unaffected and favours Lucid under both. Choosing between two deposited benchmarks is an analytical decision, which the rewrite does not make, so this is recorded rather than corrected.

The heterogeneity section places the welfare replication's conditional effects "among both the Lucid sample and respondents in the 2016 GSS," twice. The deposited welfare data hold the 1984 and 2014 General Social Surveys and no other, and the article's own Experiment 1 section names the 2014 GSS as the baseline. The deposited `standardizedExperimentsHeterogeneousEffects.RData` is a pre-computed table whose comparison sample is labelled "Original" with no year and which no deposited script builds, so which survey Figure 3's original panel draws on cannot be established from the deposit.

5 Bootstrap stability

The standard errors of appendix Table 1 come from 100 bootstrap resamples drawn without a seed, so they are different every time the deposited script is run, and so are the p-values computed from them and the counts computed from those. The point estimates are not: they involve no resampling, and all 21 reproduce to three decimals.

`appendix_table_1_bootstrap_stability.R` runs the archive's whole 100-resample procedure 200 times and records the spread. A published standard error is then checked against the sampling distribution of the estimator rather than against one draw. 20 of the 21 published standard errors fall inside the 95 per cent range of that distribution. The exception is political interest, whose published 0.082 sits just above a range ending at 0.082. One exceedance in 21 variables is what a 95 per cent interval produces by chance, and the sampler change discussed below does not widen the range enough to cover it, so the cause is left unresolved rather than guessed at.

Table 4: How much of appendix Table 1 is Monte Carlo noise: the distribution of the bootstrap standard error over 200 independent runs of the archive's own 100-resample procedure.

Variable	Published SE	Mean SE	95% interval over 200 seeds	Share of seeds significant
Age	0.032	0.032	[0.027, 0.035]	1.00
Agreeable	0.077	0.074	[0.064, 0.084]	1.00
Conscientious	0.065	0.065	[0.055, 0.075]	1.00

Table 4: How much of appendix Table 1 is Monte Carlo noise: the distribution of the bootstrap standard error over 200 independent runs of the archive's own 100-resample procedure. *(continued)*

Variable	Published SE	Mean SE	95% interval over 200 seeds	Share of seeds significant
Education	0.041	0.042	[0.037, 0.048]	1.00
Stable	0.054	0.053	[0.046, 0.060]	1.00
Extraverted	0.052	0.056	[0.048, 0.063]	0.39
Female	0.049	0.055	[0.047, 0.063]	1.00
Ideology	0.069	0.065	[0.058, 0.074]	1.00
Mean income	0.033	0.033	[0.028, 0.037]	1.00
Political Interest	0.082	0.072	[0.062, 0.082]	1.00
Open	0.069	0.065	[0.056, 0.074]	1.00
Party ID	0.055	0.055	[0.047, 0.062]	0.99
Black	0.033	0.030	[0.026, 0.034]	1.00
Hispanic	0.035	0.037	[0.031, 0.042]	0.00
White	0.041	0.041	[0.035, 0.047]	0.00
Midwest	0.046	0.045	[0.040, 0.052]	0.00
Northeast	0.054	0.051	[0.044, 0.058]	0.01
South	0.064	0.058	[0.049, 0.067]	1.00
West	0.039	0.039	[0.032, 0.044]	0.00
Voter registration	0.050	0.052	[0.045, 0.059]	0.01
Voter turnout	0.044	0.043	[0.037, 0.051]	1.00

The headline count moves too. Across 200 seeds the published figure of 14 significantly closer characteristics comes up 40% of the time, and the count ranges from 12 to 15. Under the two-tailed test it ranges from 12 to 14. A reader who runs the deposited script once and gets 13 has not found a reproducibility failure; the deposit simply does not pin the number down. The maintained rewrite sets a seed so that it does.

5.1 The sampler the archive was written under

The archive was deposited in December 2018. R 3.6.0, released in April 2019, changed how `sample()` maps a stream of uniform draws onto integers, and `rsample::bootstraps()` resamples through `sample.int()`, so the standard errors of appendix Table 1 were produced by a generator no current R installation uses by default.

Whether that matters is a question about the archive, not about the maintained rewrite, and the two are kept apart. `appendix_table_1_rounding_sampler.R` re-runs the same 200-seed procedure under `RNGkind(sample.kind = "Rounding")`, then restores the sampler explicitly and asserts the restoration, so that nothing `run_all.R` sources afterwards can inherit the old generator. The rewrite itself keeps the current sampler, which is the correct one.

Under the old sampler the published count of 14 comes up 45% of the time and ranges from 12 to 14, against 40% and 12 to 15 under the current one. The sampler change is not what makes the number move. An unseeded 100-resample bootstrap is, under either generator.

6 Ground truth

Every value in `value_paper` is transcribed from the published article or the published online appendix. Where the paper does not state a quantity, the cell is blank and the match column is empty.

Table 5: Ground truth: published value against the deposited scripts and against the maintained rewrite.

Location	Claim	Paper	Archive	Match	Rewrite	Match
Appendix Table 1	Distance, Female	0.159	0.159	1	0.159	1
Appendix Table 1	Distance, Education	0.285	0.285	1	0.285	1
Appendix Table 1	Distance, Age	0.708	0.708	1	0.708	1
Appendix Table 1	Distance, Mean income	0.099	0.099	1	0.099	1
Appendix Table 1	Distance, White	0.018	0.018	1	0.018	1
Appendix Table 1	Distance, Black	0.152	0.152	1	0.152	1
Appendix Table 1	Distance, Hispanic	-0.042	-0.042	1	-0.042	1
Appendix Table 1	Distance, Northeast	0.079	0.079	1	0.079	1
Appendix Table 1	Distance, Midwest	0.052	0.052	1	0.052	1
Appendix Table 1	Distance, South	0.137	0.137	1	0.137	1
Appendix Table 1	Distance, West	-0.007	-0.007	1	-0.007	1
Appendix Table 1	Distance, Voter registration	0.076	0.076	1	0.076	1
Appendix Table 1	Distance, Voter turnout	-0.124	-0.124	1	-0.124	1
Appendix Table 1	Distance, Party ID	0.117	0.117	1	0.117	1
Appendix Table 1	Distance, Ideology	0.465	0.465	1	0.465	1
Appendix Table 1	Distance, Political Interest	0.540	0.54	1	0.54	1
Appendix Table 1	Distance, Extraverted	0.100	0.1	1	0.1	1
Appendix Table 1	Distance, Agreeable	0.203	0.203	1	0.203	1
Appendix Table 1	Distance, Conscientious	0.420	0.42	1	0.42	1
Appendix Table 1	Distance, Stable	0.248	0.248	1	0.248	1
Appendix Table 1	Distance, Open	0.157	0.157	1	0.157	1
Appendix Table 1	Bootstrap SE, Female	0.049	0.055	1	0.055	1
Appendix Table 1	Bootstrap SE, Education	0.041	0.042	1	0.042	1
Appendix Table 1	Bootstrap SE, Age	0.032	0.032	1	0.032	1
Appendix Table 1	Bootstrap SE, Mean income	0.033	0.033	1	0.033	1
Appendix Table 1	Bootstrap SE, White	0.041	0.041	1	0.041	1
Appendix Table 1	Bootstrap SE, Black	0.033	0.03	1	0.03	1
Appendix Table 1	Bootstrap SE, Hispanic	0.035	0.036	1	0.036	1
Appendix Table 1	Bootstrap SE, Northeast	0.054	0.051	1	0.051	1
Appendix Table 1	Bootstrap SE, Midwest	0.046	0.046	1	0.046	1
Appendix Table 1	Bootstrap SE, South	0.064	0.058	1	0.058	1
Appendix Table 1	Bootstrap SE, West	0.039	0.039	1	0.039	1
Appendix Table 1	Bootstrap SE, Voter registration	0.050	0.052	1	0.052	1
Appendix Table 1	Bootstrap SE, Voter turnout	0.044	0.044	1	0.044	1
Appendix Table 1	Bootstrap SE, Party ID	0.055	0.055	1	0.055	1
Appendix Table 1	Bootstrap SE, Ideology	0.069	0.066	1	0.066	1
Appendix Table 1	Bootstrap SE, Political Interest	0.082	0.073	0	0.073	0
Appendix Table 1	Bootstrap SE, Extraverted	0.052	0.056	1	0.056	1
Appendix Table 1	Bootstrap SE, Agreeable	0.077	0.074	1	0.074	1
Appendix Table 1	Bootstrap SE, Conscientious	0.065	0.066	1	0.066	1
Appendix Table 1	Bootstrap SE, Stable	0.054	0.054	1	0.054	1
Appendix Table 1	Bootstrap SE, Open	0.069	0.065	1	0.065	1
Appendix Table 1	p-value, Female	0.000	0	1	0	1
Appendix Table 1	p-value, Education	0.000	0	1	0	1
Appendix Table 1	p-value, Age	0.000	0	1	0	1
Appendix Table 1	p-value, Mean income	0.000	0	1	0	1
Appendix Table 1	p-value, White	0.873	0.87	1	0.87	1
Appendix Table 1	p-value, Black	0.000	0	1	0	1
Appendix Table 1	p-value, Hispanic	0.357	0.383	1	0.383	1
Appendix Table 1	p-value, Northeast	0.179	0.131	1	0.131	1
Appendix Table 1	p-value, Midwest	0.409	0.398	1	0.398	1
Appendix Table 1	p-value, South	0.010	0.003	1	0.003	1
Appendix Table 1	p-value, West	0.948	0.948	1	0.948	1
Appendix Table 1	p-value, Voter registration	0.157	0.177	1	0.177	1
Appendix Table 1	p-value, Voter turnout	0.000	0	1	0	1

Table 5: Ground truth: published value against the deposited scripts and against the maintained rewrite. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
Appendix Table 1	p-value, Party ID	0.010	0.011	1	0.011	1
Appendix Table 1	p-value, Ideology	0.000	0	1	0	1
Appendix Table 1	p-value, Political Interest	0.000	0	1	0	1
Appendix Table 1	p-value, Extraverted	0.034	0.057	1	0.057	1
Appendix Table 1	p-value, Agreeable	0.001	0	1	0	1
Appendix Table 1	p-value, Conscientious	0.000	0	1	0	1
Appendix Table 1	p-value, Stable	0.000	0	1	0	1
Appendix Table 1	p-value, Open	0.005	0.002	1	0.002	1
Appendix Table 1	Two-tailed p-value, Female				0.004	
Appendix Table 1	Two-tailed p-value, Education				0	
Appendix Table 1	Two-tailed p-value, Age				0	
Appendix Table 1	Two-tailed p-value, Mean income				0.003	
Appendix Table 1	Two-tailed p-value, White				0.662	
Appendix Table 1	Two-tailed p-value, Black				0	
Appendix Table 1	Two-tailed p-value, Hispanic				0.251	
Appendix Table 1	Two-tailed p-value, Northeast				0.118	
Appendix Table 1	Two-tailed p-value, Midwest				0.259	
Appendix Table 1	Two-tailed p-value, South				0.018	
Appendix Table 1	Two-tailed p-value, West				0.85	
Appendix Table 1	Two-tailed p-value, Voter registration				0.143	
Appendix Table 1	Two-tailed p-value, Voter turnout				0.004	
Appendix Table 1	Two-tailed p-value, Party ID				0.032	
Appendix Table 1	Two-tailed p-value, Ideology				0	
Appendix Table 1	Two-tailed p-value, Political Interest				0	
Appendix Table 1	Two-tailed p-value, Extraverted				0.074	
Appendix Table 1	Two-tailed p-value, Agreeable				0.006	
Appendix Table 1	Two-tailed p-value, Conscientious				0	
Appendix Table 1	Two-tailed p-value, Stable				0	
Appendix Table 1	Two-tailed p-value, Open				0.016	
Appendix Table 2	Coefficient, lucid_base, mortalityfirst	0.898	0.898	1	0.898	1
Appendix Table 2	Coefficient, lucid_ctrl, mortalityfirst	0.898	0.898	1	0.898	1
Appendix Table 2	Coefficient, lucid_int, mortalityfirst	1.155	1.155	1	1.155	1
Appendix Table 2	Coefficient, mturk_base, mortalityfirst	1.182	1.182	1	1.182	1
Appendix Table 2	Coefficient, mturk_ctrl, mortalityfirst	1.184	1.184	1	1.184	1
Appendix Table 2	Coefficient, mturk_int, mortalityfirst	1.370	1.37	1	1.37	1
Appendix Table 2	Coefficient, kam_base, mortalityfirst	1.068	1.068	1	1.068	1
Appendix Table 2	Coefficient, kam_ctrl, mortalityfirst	1.091	1.091	1	1.091	1
Appendix Table 2	Coefficient, kam_int, mortalityfirst	1.059	1.059	1	1.059	1
Appendix Table 2	Coefficient, lucid_base, ramean	0.253	0.253	1	0.253	1
Appendix Table 2	Coefficient, lucid_ctrl, ramean	0.516	0.516	1	0.516	1
Appendix Table 2	Coefficient, lucid_int, ramean	0.522	0.522	1	0.522	1
Appendix Table 2	Coefficient, mturk_base, ramean	0.906	0.906	1	0.906	1
Appendix Table 2	Coefficient, mturk_ctrl, ramean	0.953	0.953	1	0.953	1
Appendix Table 2	Coefficient, mturk_int, ramean	1.090	1.09	1	1.09	1
Appendix Table 2	Coefficient, kam_base, ramean	0.520	0.52	1	0.52	1
Appendix Table 2	Coefficient, kam_ctrl, ramean	0.587	0.587	1	0.587	1
Appendix Table 2	Coefficient, kam_int, ramean	0.506	0.506	1	0.506	1
Appendix Table 2	Coefficient, lucid_int, mortalityfirst:ramean	-0.523	-0.523	1	-0.523	1
Appendix Table 2	Coefficient, mturk_int, mortalityfirst:ramean	-0.368	-0.368	1	-0.368	1
Appendix Table 2	Coefficient, kam_int, mortalityfirst:ramean	0.022	0.022	1	0.022	1
Appendix Table 2	Coefficient, lucid_base, (Intercept)	-0.634	-0.634	1	-0.634	1
Appendix Table 2	Coefficient, lucid_ctrl, (Intercept)	-1.040	-1.04	1	-1.04	1
Appendix Table 2	Coefficient, lucid_int, (Intercept)	-0.767	-0.767	1	-0.767	1
Appendix Table 2	Coefficient, mturk_base, (Intercept)	-1.128	-1.128	1	-1.128	1
Appendix Table 2	Coefficient, mturk_ctrl, (Intercept)	-1.193	-1.193	1	-1.193	1
Appendix Table 2	Coefficient, mturk_int, (Intercept)	-1.225	-1.225	1	-1.225	1
Appendix Table 2	Coefficient, kam_base, (Intercept)	-0.705	-0.705	1	-0.705	1
Appendix Table 2	Coefficient, kam_ctrl, (Intercept)	-0.814	-0.814	1	-0.814	1
Appendix Table 2	Coefficient, kam_int, (Intercept)	-0.700	-0.7	1	-0.7	1
Appendix Table 2	SE, lucid_base, mortalityfirst	0.065	0.065	1	0.065	1
Appendix Table 2	SE, lucid_ctrl, mortalityfirst	0.065	0.065	1	0.065	1
Appendix Table 2	SE, lucid_int, mortalityfirst	0.203	0.206	0	0.203	1

Table 5: Ground truth: published value against the deposited scripts and against the maintained rewrite. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
Appendix Table 2	SE, mturk_base, mortalityfirst	0.097	0.097	1	0.097	1
Appendix Table 2	SE, mturk_ctrl, mortalityfirst	0.098	0.098	1	0.098	1
Appendix Table 2	SE, mturk_int, mortalityfirst	0.310	0.301	0	0.31	1
Appendix Table 2	SE, kam_base, mortalityfirst	0.098	0.097	0	0.098	1
Appendix Table 2	SE, kam_ctrl, mortalityfirst	0.100	0.099	0	0.1	1
Appendix Table 2	SE, kam_int, mortalityfirst	0.294	0.294	1	0.294	1
Appendix Table 2	SE, lucid_base, ramean	0.196	0.199	0	0.196	1
Appendix Table 2	SE, lucid_ctrl, ramean	0.205	0.209	0	0.205	1
Appendix Table 2	SE, lucid_int, ramean	0.269	0.284	0	0.269	1
Appendix Table 2	SE, mturk_base, ramean	0.283	0.276	0	0.283	1
Appendix Table 2	SE, mturk_ctrl, ramean	0.293	0.29	0	0.293	1
Appendix Table 2	SE, mturk_int, ramean	0.412	0.394	0	0.412	1
Appendix Table 2	SE, kam_base, ramean	0.303	0.306	0	0.303	1
Appendix Table 2	SE, kam_ctrl, ramean	0.325	0.32	0	0.325	1
Appendix Table 2	SE, kam_int, ramean	0.485	0.482	0	0.485	1
Appendix Table 2	SE, lucid_int, mortalityfirst:ramean	0.393	0.399	0	0.393	1
Appendix Table 2	SE, mturk_int, mortalityfirst:ramean	0.569	0.554	0	0.569	1
Appendix Table 2	SE, kam_int, mortalityfirst:ramean	0.621	0.623	0	0.621	1
Appendix Table 2	SE, lucid_base, (Intercept)	0.106	0.108	0	0.106	1
Appendix Table 2	SE, lucid_ctrl, (Intercept)	0.176	0.177	0	0.176	1
Appendix Table 2	SE, lucid_int, (Intercept)	0.141	0.148	0	0.141	1
Appendix Table 2	SE, mturk_base, (Intercept)	0.163	0.161	0	0.163	1
Appendix Table 2	SE, mturk_ctrl, (Intercept)	0.281	0.28	0	0.281	1
Appendix Table 2	SE, mturk_int, (Intercept)	0.229	0.219	0	0.229	1
Appendix Table 2	SE, kam_base, (Intercept)	0.154	0.155	0	0.154	1
Appendix Table 2	SE, kam_ctrl, (Intercept)	0.275	0.271	0	0.275	1
Appendix Table 2	SE, kam_int, (Intercept)	0.229	0.227	0	0.229	1
Appendix Table 2	N, lucid_base	1629	1629	1	1629	1
Appendix Table 2	Log likelihood, lucid_base	-1027.212	-1027.212	1	-1027.212	1
Appendix Table 2	AIC, lucid_base	2060.424	2060.424	1	2060.424	1
Appendix Table 2	N, lucid_ctrl	1629	1629	1	1629	1
Appendix Table 2	Log likelihood, lucid_ctrl	-1015.762	-1015.762	1	-1015.762	1
Appendix Table 2	AIC, lucid_ctrl	2047.524	2047.524	1	2047.524	1
Appendix Table 2	N, lucid_int	1629	1629	1	1629	1
Appendix Table 2	Log likelihood, lucid_int	-1026.356	-1026.356	1	-1026.356	1
Appendix Table 2	AIC, lucid_int	2060.713	2060.713	1	2060.713	1
Appendix Table 2	N, mturk_base	766	766	1	766	1
Appendix Table 2	Log likelihood, mturk_base	-448.046	-448.046	1	-448.046	1
Appendix Table 2	AIC, mturk_base	902.092	902.092	1	902.092	1
Appendix Table 2	N, mturk_ctrl	766	766	1	766	1
Appendix Table 2	Log likelihood, mturk_ctrl	-447.878	-447.878	1	-447.878	1
Appendix Table 2	AIC, mturk_ctrl	911.755	911.755	1	911.755	1
Appendix Table 2	N, mturk_int	766	766	1	766	1
Appendix Table 2	Log likelihood, mturk_int	-447.823	-447.823	1	-447.823	1
Appendix Table 2	AIC, mturk_int	903.646	903.646	1	903.646	1
Appendix Table 2	N, kam_base	752	752	1	752	1
Appendix Table 2	Log likelihood, kam_base	-453.196	-453.196	1	-453.196	1
Appendix Table 2	AIC, kam_base	912.392	912.392	1	912.392	1
Appendix Table 2	N, kam_ctrl	752	752	1	752	1
Appendix Table 2	Log likelihood, kam_ctrl	-450.863	-450.863	1	-450.863	1
Appendix Table 2	AIC, kam_ctrl	917.727	917.727	1	917.727	1
Appendix Table 2	N, kam_int	752	752	1	752	1
Appendix Table 2	Log likelihood, kam_int	-453.195	-453.195	1	-453.195	1
Appendix Table 2	AIC, kam_int	914.391	914.391	1	914.391	1
Appendix study manifest	Welfare, GSS 1984 N	943	943	1	943	1
Appendix study manifest	Welfare, GSS 2014 N	2457	2457	1	2457	1
Appendix study manifest	Welfare, Lucid N	1811	3294	0	3294	0
Appendix study manifest	Welfare, MTurk N	494	494	1	494	1
Appendix study manifest	Asian Disease, original N	307	307	1	307	1
Appendix study manifest	Asian Disease, Lucid N	1813	1813	1	1813	1
Appendix study manifest	Kam and Simas, original N	761	752	0	752	0
Appendix study manifest	Kam and Simas, Lucid N	1629	1629	1	1629	1
Appendix study manifest	Kam and Simas, MTurk N	766	766	1	766	1
Appendix study manifest	Hiscox, original N	1578	1578	1	1578	1
Appendix study manifest	Hiscox, Lucid N	1811	1811	1	1811	1
Appendix study manifest	Hiscox, GfK N	2084	2084	1	2084	1

Table 5: Ground truth: published value against the deposited scripts and against the maintained rewrite. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
Appendix study manifest	Hiscox, MTurk N	2972	2972	1	2972	1
Appendix study manifest	Healthcare rumors, original N	1593	1593	1	1593	1
Appendix study manifest	Healthcare rumors, Lucid N	3503	3503	1	3503	1
Appendix study manifest	Hiscox, GfK wave 2 N	1838				
Appendix study manifest	Hiscox, MTurk wave 2 N	2307				
Text, Our sample	Lucid sample N	3504	3504	1	3504	1
Text, Our sample	Surveys per month, mean	4.28	4.28	1	4.28	1
Text, Our sample	Percent taking fewer than one survey per day	98	98.06	1	98.06	1
Text, Our sample	Surveys per month among those, mean	2.43	2.43	1	2.43	1
Text, Our sample	Percent taking surveys at home	94	93.51	1	93.51	1
Text, Our sample	Expected compensation, mean dollars	5.01	5.01	1	5.01	1
Text, Our sample	Expected compensation, mean dollars, trimmed	1.16	1.16	1	1.16	1
Text, Baseline characteristics	Percent female, Lucid	52	52.15	1	52.15	1
Text, Baseline characteristics	Percent female, MTurk	60	60.07	1	60.07	1
Text, Baseline characteristics	Years of education, Lucid	14.2	14.16	1	14.16	1
Text, Baseline characteristics	Years of education, ANES	13.5	13.5	1	13.5	1
Text, Baseline characteristics	Years of education, MTurk		14.88		14.88	
Text, Baseline characteristics	Party ID, Lucid	3.7	3.73	1	3.73	1
Text, Baseline characteristics	Party ID, ANES 2012	3.7	3.73	1	3.73	1
Text, Baseline characteristics	Party ID, MTurk	3.5	3.49	1	3.49	1
Text, Baseline characteristics	Political interest, Lucid minus MTurk	1.2	1.19	1	1.19	1
Text, Baseline characteristics	Political knowledge, Lucid		58.39		58.39	
Text, Baseline characteristics	Political knowledge, MTurk		70.51		70.51	
Text, Baseline characteristics	Political knowledge, ANES panel		59.91		59.91	
Text, Distance tests	Variables tested	21	21	1	21	1
Text, Distance tests	Lucid closer	18	18	1	18	1
Text, Distance tests	MTurk closer	3	3	1	3	1
Text, Distance tests	Lucid significantly closer, archive p	14	13	0	13	0
Text, Distance tests	MTurk significantly closer, archive p	1	1	1	1	1
Text, Distance tests	Lucid significantly closer, two-tailed p		13		13	
Text, Distance tests	MTurk significantly closer, two-tailed p		1		1	
Text, Distance tests	Demographic variables tested	11	11	1	11	1
Text, Distance tests	Demographic, Lucid closer	9	9	1	9	1
Text, Distance tests	Demographic, significantly closer, archive p	5	6	0	6	0
Text, Distance tests	Political, significantly closer, archive p	3	3	1	3	1
Text, Distance tests	Traits, significantly closer, archive p	5	4	0	4	0
Text, Distance tests	Traits, significantly closer, two-tailed p		4		4	
Text, Healthcare rumors	Death panel belief, Lucid control mean	-0.17	-0.172	1	-0.172	1
Text, Healthcare rumors	Death panel belief, original control mean	-0.19	-0.19	1	-0.19	1
Text, Our sample, p.4	share paid in dollars or points exceeds half		1		1	
Text, Baseline characteristics, p.5	gender, education, age and income all favour Lucid		1		1	
Text, Baseline characteristics, p.5	education distance favours Lucid		1		1	
Text, Baseline characteristics, p.5	mean and median income both sit between MTurk and the ANES 2012 face-to-face sample		1		1	
Text, Baseline characteristics, p.5	both samples sit above the benchmark white share and Lucid by less		1		1	
Text, Baseline characteristics, p.5	MTurk's southern share sits below the benchmark, not above it		0		0	
Text, Baseline characteristics, p.6	registration favours Lucid and turnout favours MTurk		0		0	
Text, Baseline characteristics, p.6	MTurk sits furthest below the benchmark on the conservatism scale		1		1	
Text, Baseline characteristics, p.6	MTurk lowest and Lucid highest on political interest		1		1	
Text, Baseline characteristics, p.6	MTurk above the ANES panel and Lucid nearer to it than MTurk is		1		1	
Text, Baseline characteristics, p.6	three characteristics favour Lucid and turnout favours MTurk, all significant		1		1	

Table 5: Ground truth: published value against the deposited scripts and against the maintained rewrite. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
Text, Baseline characteristics, p.6	MTurk is lowest on support for senior drug benefits and youngest		1		1	
Text, Baseline characteristics, p.6	Lucid nearer the ANES 2012 than MTurk on conscientiousness and stability		1		1	
Text, Baseline characteristics, p.6	under the two-tailed test the appendix describes, extraversion does not clear 0.05		0		0	
Text, Experiments, p.6	the Berinsky facet carries no MTurk estimate		1		1	
Text, Treatment effect heterogeneity, p.9	the three race conditional effects overlap in both samples		1		1	
Text, Treatment effect heterogeneity, p.9	the risk acceptance interaction spans zero in both samples		1		1	
Text, Treatment effect heterogeneity, p.9	conditional effects overlap within every frame by sample cell		1		1	
Text, Treatment effect heterogeneity, p.9	the Democratic correction among Democrats is the only conditional effect that differs between samples		1		1	
Text, Discussion, p.12	more than half the Lucid estimates agree with the original on sign and significance		1		1	
Text, Appendix 1	the published p-value column is the archive's own expression, not the two-tailed test		0		0	
Text, Treatment effect heterogeneity, p.9	experiments with at least one conditional average treatment effect	4	4	1	4	1
Text, Baseline characteristics, p.6	policy items with an unambiguous direction on which Lucid polls more conservatively		2		2	
Text, Discussion, p.12	Lucid estimates that are significant with the opposite sign from the original	0	0	1	0	1
Text, Discussion, p.12	experiments with at least one sign or significance mismatch against the original		2		2	
Text, Fit for purpose	Experiments replicated	5	5	1	5	1
Figure 1	ANES 2012 standardized to mean zero		4.74200754701305e-16		4.74200754701305e-16	
Text, Experiments	Confidence level	95	95.0598242256987	1	95.0598242256987	1
Text, Experiment 4	Levels of the expert factor	2	2	1	2	1
Text, Experiment 4	Levels of the valence factor	4	4	1	4	1
Text, Treatment effect heterogeneity	GSS baseline year	2016	2014	0	2014	0
Figure 1	All plotted estimates and confidence bounds					
Figure 2	All plotted estimates and confidence bounds					
Figure 3	All plotted estimates and confidence bounds					
Figure 4	All plotted estimates and confidence bounds					
Figure 5	All plotted estimates and confidence bounds					
Table 1	Theoretical applicability of five experimental studies					

Of the 259 recorded claims, 199 can be compared against a published value. 167 of those match what the deposited scripts produce and 32 do not; 192 match the maintained rewrite and 7 do not.

7 Maintained rewrite

The rewrite lives in `maintained/`: seventeen scripts and a `helpers.R`, covering the nine tables the archive's README names, the five figures, the two published appendix tables, two stability checks on the bootstrap, and the article's in-text numbers. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used.

Table 6: Maintained rewrite scripts, the archive script each replaces, and what of it the published paper actually shows.

Script	Archive script	Where it reaches print
helpers.R	LucidValidationHelperFunctions.R	
table_1_demographics.R	01	Figure 1 only
table_2_political.R	02	Figure 1 and prose
table_3_psychological.R	03	Figure 1
table_4_welfare.R	04	Figure 2
table_5_asian_disease.R	05	Figure 2
table_6_kamsimas.R	06 (OLS half)	Figure 2
table_7_hiscox.R	07	Figure 2
table_8_health_rumors.R	08	Figure 2 and prose
table_9_survey_behavior.R	09	prose
figure_1_standardized_demos.R	10	Figure 1
figure_2_standardized_experiments.R	10	Figure 2
figures_3_4_5_heterogeneous_effects.R	11	Figures 3 to 5
appendix_table_1_distance_tests.R	12	Appendix Table 1
appendix_table_1_bootstrap_stability.R	none	
appendix_table_1_rounding_sampler.R	none	
appendix_table_2_kamsimas_probit.R	06 (probit half)	Appendix Table 2
text_in_text_claims.R	none	prose

7.1 Deprecated patterns replaced

Table 7: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
magrittr pipe and ‘.‘ functional sequences	native pipe
‘summarize_all(funs(...))‘	‘summarize(across(...))‘
‘reshape2::melt‘ / ‘dcast‘	‘pivot_longer()‘ / ‘pivot_wider()‘
‘within()‘ blocks over base indexing	‘mutate()‘ paragraphs
‘lm()‘ + ‘starprep()‘ + ‘stargazer‘	‘estimatr::lm_robust()‘, whose default is HC2, plus ‘modelsummary(output =)‘
‘glm()‘ + ‘stargazer‘ with no ‘se =‘	‘glm()‘ + ‘sandwich::vcovHC(type = ”HC2”)‘, both SEs written out
‘sink()‘ to a hardcoded path	‘write_csv()‘ to ‘output/‘
‘load(”*.rdata”)‘ against a deposited ‘.RData‘	the name the deposit actually uses, checked against ‘ls‘ output
‘coefplot::position_dodgev(height =)‘	‘position_dodge(width = 0.5)‘
‘geom_errorbarh(height = 0)‘	‘geom_linerange()‘

‘unnest()‘ with no ‘cols =‘	‘unnest(cols = c(model))‘
‘weighted.se()‘ from the archive’s helper file	‘weighted_se_mean()‘ in ‘helpers.R‘, same formula
positional assignment of display names	named lookup keyed on the variable name
unseeded bootstrap	‘set.seed(20260321)‘, plus a stability check over 200 seeds

Two rows of that table describe intentions rather than live code. `xtable` is loaded by five archive scripts and called by none of them: every `xtable()` and `print.xtable()` sits inside a commented-out block. The other five, the ones that build regression tables, bracket their `stargazer` calls in a commented `sink("...tex")` and `sink()` pair, so the `stargazer` call is live and the redirection that would have captured it is not. The six `.tex` files those `sink()` lines name exist nowhere in the archive. The rewrite writes every table to `maintained/output/` as CSV, and the regression tables additionally as `.tex` through `modelsummary(output =)`, which needs no `sink()` and so cannot be disabled by commenting one line out.

8 Figures

The five figures are the paper’s main quantitative content. Each is drawn from a data file the archive deposits pre-computed, so the rewrite redraws them rather than recomputing their contents.

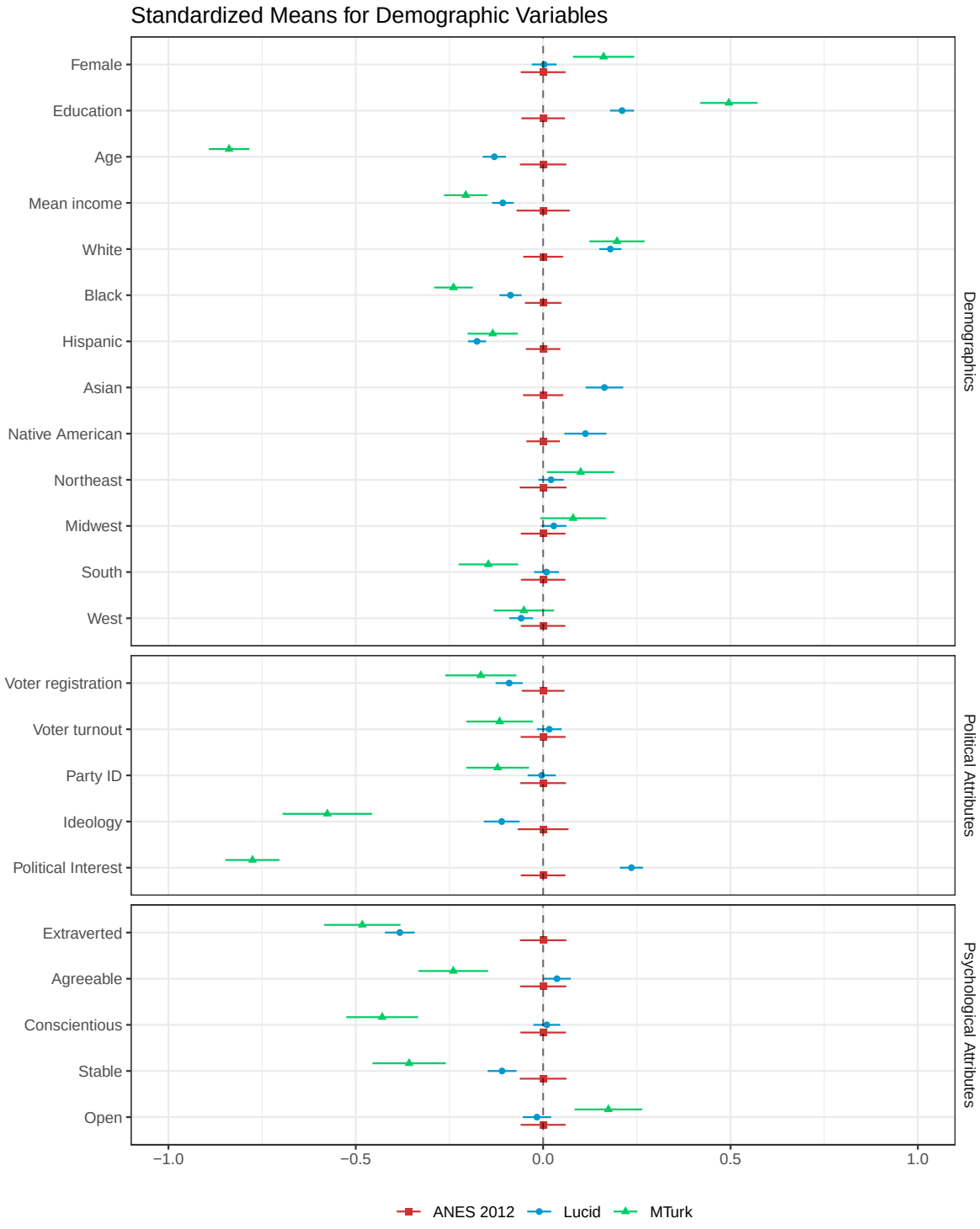


Figure 1: Figure 1, standardized means for demographic, political and psychological variables, as reproduced by the maintained rewrite.

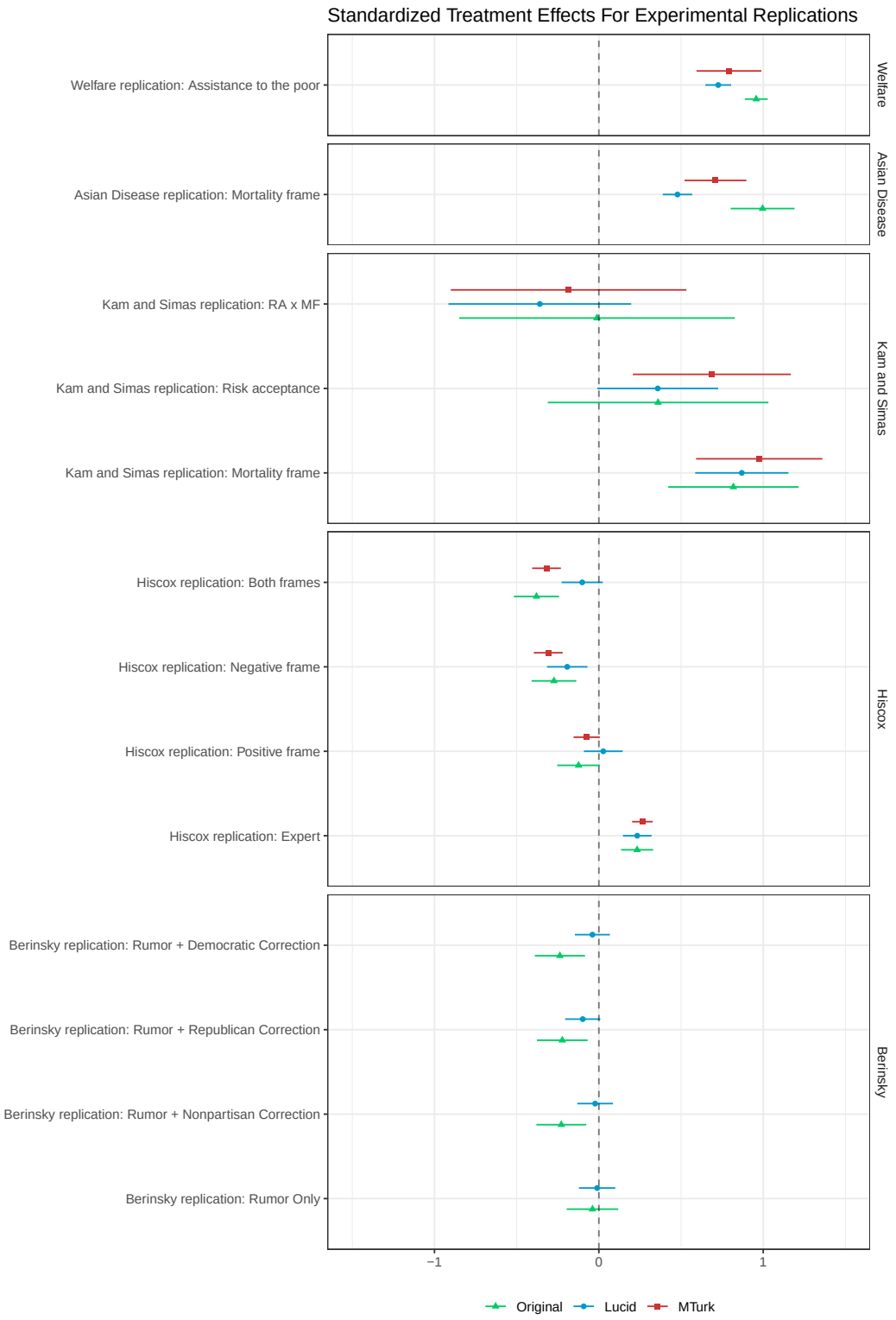


Figure 2: Figure 2, standardized treatment effects for the five experimental replications, as reproduced by the maintained rewrite.

9 In-text numbers

The article states most of its quantitative content in prose rather than in tables, so `maintained/text_in_text_claims.R` is where the bulk of the number-by-number check happens. That script reads only from `maintained/output/` and holds no published value at all. The article's own figures enter one file, `ground_truth/build_ground_truth.R`, and the two are joined there, which is what the table below shows.

Table 8: In-text claims. A blank article column marks a quantity the pipeline produces to support a comparative claim the article states in words rather than in numbers.

Section	Claim	Pipeline	Article
Our sample	Lucid sample N	3504	3504
Our sample	Surveys per month, mean	4.28	4.28
Our sample	Percent taking fewer than one survey per day	98.06	98
Our sample	Surveys per month among those, mean	2.43	2.43
Our sample	Percent taking surveys at home	93.51	94
Our sample	Expected compensation, mean dollars	5.01	5.01
Our sample	Expected compensation, mean dollars, trimmed	1.16	1.16
Baseline characteristics	Percent female, Lucid	52.15	52
Baseline characteristics	Percent female, MTurk	60.07	60
Baseline characteristics	Years of education, Lucid	14.16	14.2
Baseline characteristics	Years of education, ANES	13.5	13.5
Baseline characteristics	Years of education, MTurk	14.88	
Baseline characteristics	Party ID, Lucid	3.73	3.7
Baseline characteristics	Party ID, ANES 2012	3.73	3.7
Baseline characteristics	Party ID, MTurk	3.49	3.5
Baseline characteristics	Political interest, Lucid minus MTurk	1.19	1.2
Baseline characteristics	Political knowledge, Lucid	58.39	
Baseline characteristics	Political knowledge, MTurk	70.51	
Baseline characteristics	Political knowledge, ANES panel	59.91	
Distance tests	Variables tested	21	21
Distance tests	Lucid closer	18	18
Distance tests	MTurk closer	3	3
Distance tests	Lucid significantly closer, archive p	13	14
Distance tests	MTurk significantly closer, archive p	1	1
Distance tests	Lucid significantly closer, two-tailed p	13	
Distance tests	MTurk significantly closer, two-tailed p	1	
Distance tests	Demographic variables tested	11	11
Distance tests	Demographic, Lucid closer	9	9
Distance tests	Demographic, significantly closer, archive p	6	5
Distance tests	Political, significantly closer, archive p	3	3
Distance tests	Traits, significantly closer, archive p	4	5
Distance tests	Traits, significantly closer, two-tailed p	4	
Healthcare rumors	Death panel belief, Lucid control mean	-0.172	-0.17
Healthcare rumors	Death panel belief, original control mean	-0.19	-0.19
Our sample, p.4	share paid in dollars or points exceeds half	1	
Baseline characteristics, p.5	gender, education, age and income all favour Lucid	1	
Baseline characteristics, p.5	education distance favours Lucid	1	
Baseline characteristics, p.5	mean and median income both sit between MTurk and the ANES 2012 face-to-face sample	1	

Table 8: In-text claims. A blank article column marks a quantity the pipeline produces to support a comparative claim the article states in words rather than in numbers. *(continued)*

Section	Claim	Pipeline	Article
Baseline characteristics, p.5	both samples sit above the benchmark white share and Lucid by less	1	
Baseline characteristics, p.5	MTurk's southern share sits below the benchmark, not above it	0	
Baseline characteristics, p.6	registration favours Lucid and turnout favours MTurk	0	
Baseline characteristics, p.6	MTurk sits furthest below the benchmark on the conservatism scale	1	
Baseline characteristics, p.6	MTurk lowest and Lucid highest on political interest	1	
Baseline characteristics, p.6	MTurk above the ANES panel and Lucid nearer to it than MTurk is	1	
Baseline characteristics, p.6	three characteristics favour Lucid and turnout favours MTurk, all significant	1	
Baseline characteristics, p.6	MTurk is lowest on support for senior drug benefits and youngest	1	
Baseline characteristics, p.6	Lucid nearer the ANES 2012 than MTurk on conscientiousness and stability	1	
Baseline characteristics, p.6	under the two-tailed test the appendix describes, extraversion does not clear 0.05	0	
Experiments, p.6	the Berinsky facet carries no MTurk estimate	1	
Treatment effect heterogeneity, p.9	the three race conditional effects overlap in both samples	1	
Treatment effect heterogeneity, p.9	the risk acceptance interaction spans zero in both samples	1	
Treatment effect heterogeneity, p.9	conditional effects overlap within every frame by sample cell	1	
Treatment effect heterogeneity, p.9	the Democratic correction among Democrats is the only conditional effect that differs between samples	1	
Discussion, p.12	more than half the Lucid estimates agree with the original on sign and significance	1	
Appendix 1	the published p-value column is the archive's own expression, not the two-tailed test	0	
Treatment effect heterogeneity, p.9	experiments with at least one conditional average treatment effect	4	4
Baseline characteristics, p.6	policy items with an unambiguous direction on which Lucid polls more conservatively	2	
Discussion, p.12	Lucid estimates that are significant with the opposite sign from the original	0	0
Discussion, p.12	experiments with at least one sign or significance mismatch against the original	2	
Fit for purpose	Experiments replicated	5	5
Experiments	Confidence level	95.0598242256987	95
Experiment 4	Levels of the expert factor	2	2
Experiment 4	Levels of the valence factor	4	4
Treatment effect heterogeneity	GSS baseline year	2014	2016

The one benchmark this cannot check is the 50.8% Census female share the article compares its 52% against: it is cited from outside the deposit and no file here contains it.

9.1 Coverage, and the second instrument

The table above shows what the pipeline produces. What decides whether that is enough is `ground_truth/published_claims.csv`, an extraction of every quantity the article and the online appendix state, obtained by pulling every numeric token out of both documents with its surrounding context, sweeping separately for numbers spelled out as words, and classifying each by hand. It holds 255 claims. 221 of them are quantities the pipeline can reach and each has a block in `maintained/in_text_claims.R`; the rest are scale endpoints, question wordings, field dates and figures cited from other papers, which nothing here can check and which are marked as such rather than passed over.

25 of the claims are descriptive: sentences about shape, sign or count, with no number in them at all. "The MTurk sample appears to overrepresent southerners" and "Lucid is significantly closer to the ANES 2012 than MTurk on all five traits" are of that kind, and neither would be caught by any comparison of numbers, because neither states one. `maintained/text_descriptive_claims.R` gives each a computed truth value from the estimates in `maintained/output/`, and the two that come back false are in the errata.

`maintained/in_text_claims.R` prints every claim beside the sentence that makes it. It reaches each number out of `maintained/output/` by its own path, doing its own selection and rounding, and it never reads the ground truth. `build_ground_truth.R` runs it, counts the claims it printed against the extraction in both directions, and compares the two instruments value by value; a disagreement between them stops the build, as does a claim with a block and no ground-truth row, or a row and no block.

10 Rewrite verification

Two consecutive runs of `run_all.R` produce byte-identical CSV, TeX and PNG output. The five PDF figures differ between runs and only between runs, because a PDF records the time it was written; their PNG twins are the copies to compare. Every stochastic script sets a seed, and the two that sweep seeds deliberately sweep a fixed range.

No maintained script takes a published number as an input. Every value in `maintained/output/` traces to the deposited data, and the published figures appear only in `ground_truth/build_ground_truth.R`, where they are transcribed from the article and the online appendix and used only as comparison targets. `build_ground_truth.R` reads `value_rewrite` back out of `maintained/output/` rather than carrying a typed copy, so the ground truth cannot drift from the scripts, and it runs as the last step of `run_all.R`.

The deposit itself is checked rather than trusted. `download_original.R` verifies all 29 files against the MD5 of the bytes Dataverse serves, stops if any disagrees, and also stops if `original/` holds a single file the manifest does not list, since a checksum can only speak for files it knows about. It runs at the top of `run_all.R`, so the check happens on every run rather

than once. The repository holds no copy of the deposit, and the archive's own scripts are run from a scratch copy, so a script that writes into its working directory cannot alter the deposited bytes. Running all twelve numbered scripts leaves the copy byte-identical to the deposit, which is a consequence of the archive writing nothing at all.

11 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-09-27

Table 10: Package versions used for the run behind this report.

Package	Version
tidyverse	2.0.0
estimatr	2.0.1
broom	1.0.13
scales	1.4.0
matrixStats	1.5.0
rsample	1.3.2
sandwich	3.1.1
modelsummary	2.6.0
ggplot2	4.0.3
here	1.0.2